

Multi-Container Orchestration of Spring Boot and PostgreSQL Using Docker Compose

```
Command Prompt
Directory of D:\postgre-springboot-docker
12-05-2026 11:14 <DIR> .
12-05-2026 11:34 1,030 docker-compose.yml
12-05-2026 11:42 <DIR> spring-app
1 File(s) 1,030 bytes
2 Dir(s) 137,631,961,088 bytes free

D:\postgre-springboot-docker>cd spring-app

D:\postgre-springboot-docker\spring-app>dir
Volume in drive D is New Volume
Volume Serial Number is AEE6-4A19

Directory of D:\postgre-springboot-docker\spring-app
12-05-2026 11:42 <DIR> .
12-05-2026 11:14 <DIR> ..
12-05-2026 10:55 38 .gitattributes
12-05-2026 10:55 394 .gitignore
12-05-2026 10:55 <DIR> .mvn
12-05-2026 11:05 136 Dockerfile
12-05-2026 10:55 1,447 HELP.md
12-05-2026 10:55 11,790 mvnw
12-05-2026 10:55 8,292 mvnw.cmd
12-05-2026 10:55 1,479 pom.xml
12-05-2026 10:55 <DIR> src
12-05-2026 11:42 <DIR> target
7 File(s) 23,576 bytes
5 Dir(s) 137,631,961,088 bytes free
```

start.spring.io/?utm_source=chatgpt.com

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer All Bookmarks

spring initializr

Project

☐ Gradle - Groovy ☒ **Java** ☐ Kotlin ☐ Groovy

☐ Gradle - Kotlin ☒ **Maven**

Spring Boot

☐ 4.1.0 (SNAPSHOT) ☐ 4.1.0 (RC1) ☐ 4.0.7 (SNAPSHOT) ☐ 4.0.6

☐ 3.5.15 (SNAPSHOT) ☒ **3.5.14**

Project Metadata

Group

Artifact

Package name

Packaging ☒ **Jar** ☐ War

Configuration ☒ **Properties** ☐ YAML

Java ☐ 26 ☐ 25 ☐ 21 ☒ **17**

Dependencies ADD DEPENDENCIES... CTRL + B

Spring Web WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

PostgreSQL Driver SQL

A JDBC and R2DBC driver that allows Java programs to connect to a PostgreSQL database using standard, database independent Java code.

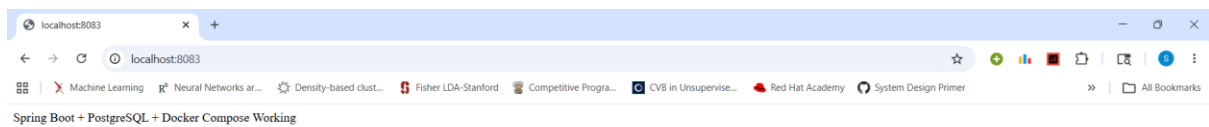
GENERATE CTRL + G EXPLORE CTRL + SPACE ...

```
Command Prompt - mvnw.cnr x + v
2026-05-12T13:39:33.606+05:30 INFO 20508 --- [spring-app] [main] c.e.s.SpringAppApplicationTests :
No active profile set, falling back to 1 default profile: "default"
2026-05-12T13:39:35.217+05:30 INFO 20508 --- [spring-app] [main] c.e.s.SpringAppApplicationTests :
Started SpringAppApplicationTests in 2.01 seconds (process running for 3.287)
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been
appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.337 s -- in com.example.spring_app.SpringAppApp
licationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ spring-app ---
[INFO] Building jar: D:\postgre-springboot-docker\spring-app\target\spring-app-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:3.5.14:repackage (repackage) @ spring-app ---
[INFO] Replacing main artifact D:\postgre-springboot-docker\spring-app\target\spring-app-0.0.1-SNAPSHOT.jar with repacka
ged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to D:\postgre-springboot-docker\spring-app\target\spring-app-0.0.1-SNAPSHO
T.jar.original
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.794 s
[INFO] Finished at: 2026-05-12T13:39:37+05:30
[INFO] -----
D:\postgre-springboot-docker\spring-app>
```

```
Command Prompt - mvnw.cnr x + v
#9 exporting manifest sha256:e5b17b413e940fce26b251afcd5c7ccfba50f331a982cfc82bbc0a296e6856df 0.0s done
#9 exporting config sha256:79d72c0eca79a5a42736817a77926c727153834d9b5ab5488b98e068e6cc892c 0.0s done
#9 exporting attestation manifest sha256:baf4be3df80ebb0ad81c0a57a74872c48e26d60e0499a314743e2868b2630471 0.0s done
#9 exporting manifest list sha256:f928a7f41979deacd9bd1a6e13637e5552e1be189cc51c4c2c1ca5cc30b7e6bd
#9 exporting manifest list sha256:f928a7f41979deacd9bd1a6e13637e5552e1be189cc51c4c2c1ca5cc30b7e6bd 0.0s done
#9 naming to docker.io/library/postgre-springboot-docker-springboot-app:latest done
#9 unpacking to docker.io/library/postgre-springboot-docker-springboot-app:latest
#9 unpacking to docker.io/library/postgre-springboot-docker-springboot-app:latest 0.3s done
#9 DONE 1.3s

#10 resolving provenance for metadata file
#10 DONE 0.0s
[+] up 2/2
✔Image postgre-springboot-docker-springboot-app Built 4.2s
✔Container springboot-app-container Recreated 0.3s
Attaching to postgres-db, springboot-app-container
Container postgres-db Waiting
postgres-db |
postgres-db | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres-db |
postgres-db | 2026-05-12 08:11:29.976 UTC [1] LOG: starting PostgreSQL 17.9 (Debian 17.9-1.pgdg13+1) on x86_64-pc-linu
x-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
postgres-db | 2026-05-12 08:11:29.977 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
postgres-db | 2026-05-12 08:11:29.977 UTC [1] LOG: listening on IPv6 address ":::", port 5432
postgres-db | 2026-05-12 08:11:29.985 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres-db | 2026-05-12 08:11:29.995 UTC [27] LOG: database system was shut down at 2026-05-12 06:13:54 UTC
postgres-db | 2026-05-12 08:11:30.015 UTC [1] LOG: database system is ready to accept connections
```

```
Command Prompt - mvnw.cr x + v
Container postgres-db Healthy
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
springboot-app-container
Application : Starting SpringAppApplication v0.0.1-SNAPSHOT using Java 17.0.18 with PID 1 (/app/app.jar started by
root in /app)
springboot-app-container | 2026-05-12T08:11:37.461Z INFO 1 --- [spring-app] [ main] c.e.spring_app.SpringApp
Application : No active profile set, falling back to 1 default profile: "default"
springboot-app-container | 2026-05-12T08:11:37.469Z INFO 1 --- [spring-app] [ main] c.e.spring_app.SpringApp
springboot-app-container | 2026-05-12T08:11:38.834Z INFO 1 --- [spring-app] [ main] o.s.b.w.embedded.tomcat.
TomcatWebServer : Tomcat initialized with port 8080 (http)
springboot-app-container | 2026-05-12T08:11:38.855Z INFO 1 --- [spring-app] [ main] o.apache.catalina.core.S
standardService : Starting service [Tomcat]
springboot-app-container | 2026-05-12T08:11:38.855Z INFO 1 --- [spring-app] [ main] o.apache.catalina.core.S
standardEngine : Starting Servlet engine: [Apache Tomcat/10.1.54]
springboot-app-container | 2026-05-12T08:11:38.893Z INFO 1 --- [spring-app] [ main] o.a.c.c.C.[Tomcat].[loca
lhost].[/] : Initializing Spring embedded WebApplicationContext
springboot-app-container | 2026-05-12T08:11:38.896Z INFO 1 --- [spring-app] [ main] w.s.c.ServletWebServerAp
plicationContext : Root WebApplicationContext: initialization completed in 1327 ms
springboot-app-container | 2026-05-12T08:11:39.516Z INFO 1 --- [spring-app] [ main] o.s.b.w.embedded.tomcat.
TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
springboot-app-container | 2026-05-12T08:11:39.548Z INFO 1 --- [spring-app] [ main] c.e.spring_app.SpringApp
Application : Started SpringAppApplication in 2.906 seconds (process running for 3.83)
```



```
Command Prompt - mvnw.cr x + v
services:
  springboot-app:
    build:
      context: ./spring-app
      container_name: springboot-app-container
    ports:
      - "8083:8080"
    environment:
      SPRING_DATASOURCE_URL: jdbc:postgresql://postgres:5432/my_db
      SPRING_DATASOURCE_USERNAME: myuser
      SPRING_DATASOURCE_PASSWORD: mypassword
    depends_on:
      postgres:
        condition: service_healthy
    restart: unless-stopped
    networks:
      - backend-network

  postgres:
    image: postgres:17
    container_name: postgres-db
    ports:
      - "5432:5432"
    environment:
      POSTGRES_DB: my_db
      POSTGRES_USER: myuser
      POSTGRES_PASSWORD: mypassword
    volumes:
      - postgres_data:/var/lib/postgresql/data
    networks:
```

```
Command Prompt - mvnw.cr  x  +  v

container_name: postgres-db
ports:
  - "5432:5432"
environment:
  POSTGRES_DB: my_db
  POSTGRES_USER: myuser
  POSTGRES_PASSWORD: mypassword
volumes:
  - postgres_data:/var/lib/postgresql/data
networks:
  - backend-network
restart: unless-stopped
healthcheck:
  test: ["CMD-SHELL", "pg_isready -U myuser -d my_db"]
  interval: 10s
  timeout: 3s
  retries: 5
  start_period: 10s
volumes:
  postgres_data:
networks:
  backend-network:

D:\postgre-springboot-docker>type spring-app\dockerfile
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/spring-app-0.0.1-SNAPSHOT.jar app.jar
EXPOSE 8080
CMD ["java", "-jar", "app.jar"]
D:\postgre-springboot-docker>
```

```
Command Prompt - mvnw.cr  x  +  v

container_name: postgres-db
ports:
  - "5432:5432"
environment:
  POSTGRES_DB: my_db
  POSTGRES_USER: myuser
  POSTGRES_PASSWORD: mypassword
volumes:
  - postgres_data:/var/lib/postgresql/data
networks:
  - backend-network
restart: unless-stopped
healthcheck:
  test: ["CMD-SHELL", "pg_isready -U myuser -d my_db"]
  interval: 10s
  timeout: 3s
  retries: 5
  start_period: 10s
volumes:
  postgres_data:
networks:
  backend-network:

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FROM eclipse-temurin:17-jdk
WORKDIR /app
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